

SIF-Sense AI: Complete Technical & User Guide

A-to-Z Manual: OSHA 2015–2025 AI Model, Dataset Science, Page-by-Page Walkthrough & Free Database Storage Architecture

Target System: SIF-Sense AI Early Warning System	Database: MongoDB Atlas M0 (Free Tier 512MB) + SQLite
AI Model: Calibrated OSHA 2015–2025 Model (106k reports)	User Account: kanishkskcet_db_user

SECTION 1: What AI/ML Model & What Data Are We Using?

1. What is the Dataset?

We train directly on the official **United States Occupational Safety and Health Administration (OSHA) Severe Injury Reports dataset** spanning exactly 10 years from **January 2015 to November 2025** (105,996 total records). By federal law (OSHA Rule 29 CFR 1904.39), all US employers must report any work-related in-patient hospitalization, amputation, or loss of an eye within 24 hours. Key attributes include: *Final Narrative* (detailed description of what happened), *Hospitalized count*, *Amputation count*, *Loss of Eye count*, *Nature of Injury* (Fractures, Burns, Crush, Trauma), *Part of Body*, *Event mechanism* (Caught in machinery, Fall from height, Arc flash), and *Industry NAICS code*.

2. What Exact AI/ML Model Are We Using?

Rather than slow generic cloud models that incur per-call API fees and privacy concerns, SIF-Sense AI deploys a **three-tier ultra-fast machine learning pipeline**:

- **Text Representation (TF-IDF N-grams):** Converts natural language text into 25,000 sublinear TF-IDF features capturing single words, bigrams, and trigrams (1-3 n-grams) like 'lockout tagout', 'fall from height', 'conveyor nip point', and 'arc flash'.
- **Calibrated SIF Classifier (SGD + Platt Sigmoid Scaling):** A high-performance linear model trained via Stochastic Gradient Descent with log-loss and balanced class weighting, wrapped in a 3-fold Platt Sigmoid calibrator. This outputs real, mathematically sound probabilities (0.00% to 100.0%) rather than arbitrary raw scores.
- **Multi-Class Injury Nature Classifier:** A companion multi-nominal logistic regression classifier that predicts the physical trauma category (Amputation, Fracture, Crushing Injury, Thermal/Chemical Burn, Head/Brain Trauma, Electrical Trauma, or Laceration).
- **Heuristic Safety Guardrails:** A rule-based expert safety engine that acts as a failsafe to guarantee that catastrophic zero-tolerance indicators (high voltage, toxic H2S, crane collapse) are never missed.

Metric	Result	Meaning in Plain English
Trained Records	105,996 rows	84,796 train / 21,199 unseen test split
Model Accuracy	96.00%	Overall correct classifications on 21,199 test cases
SIF Safety Recall	97.49%	Crucial: 97.5% of real severe life-altering threats are caught
Precision	97.18%	Extremely low false alarm rate
F1-Score	0.9733	Harmonic balance between safety recall and accuracy
Inference Speed	< 15 ms	Ultra-fast sub-second execution on standard CPU (zero GPU cost)

SECTION 2: Managing Storage on Free MongoDB Atlas (512 MB Limit)

MongoDB Atlas Free Tier (M0) provides exactly **512 MB of total shared storage**. To prevent your database from filling up while maintaining full safety compliance, SIF-Sense AI implements a **4-pillar storage solution**:

- **1. Automated TTL (Time-To-Live) Expiration:** Non-critical housekeeping logs and dismissed observations automatically delete after 90 days. Genuine SIF precursors are permanently retained with zero risk of deletion:

```
db.reports.create_index(
    [{"created_at": 1}],
    expireAfterSeconds=7776000, # 90 days
    partialFilterExpression={"sif_potential": False} # Never deletes SIF events!
)
```

- **2. Document Schema Pruning:** Bulky debug logs and raw embeddings are stripped out in mongo_manager.py. Documents are compressed to **< 380 bytes**, allowing 512 MB to hold **1,340,000+ reports**.
- **3. 1-Click Offline Archival:** The Report Log has built-in CSV & JSON export so safety directors can download logs to cold storage and flush Atlas at any time.

SECTION 3: Complete Page-by-Page User Manual

Page 1: SIF Incident Analyzer (URL: /)

The primary AI evaluation console where safety observations and incident descriptions are analyzed in real time.

- **Adaptive Layout:** Side-by-side on desktop; smooth two-tab flow (1. *Describe Incident* & 2. *Safety Result*) on mobile/narrow viewports.
- **1-Click Real OSHA Benchmark Presets:** Pre-loaded with authentic federal severe injury incidents: Machinery Entanglement, Fall from Height, Arc Flash Burn, Suspended Load Crush, Confined Space Toxic Gas, and Minor Office Scratch.
- **Instant Calibrated Priority Score:** Circular gauge scoring risk from 0 to 100 based on severity, exposure, barrier failure, and historical recurrence.
- **Word-Level Risk Highlights:** Highlights exact trigger words learned from 106,000 OSHA records (e.g., *amputation* (+21.0), *fingers* (+4.6), *energized* (+2.4)).
- **Predicted Injury Nature:** Classifies likely physical trauma (Amputation, Fracture, Crushing Injury, Thermal/Chemical Burn).
- **IOGP Life-Saving Rules:** Recommends immediate plain-English mitigation actions.

Page 2: HSE Command Center Dashboard (URL: /dashboard)

Executive oversight dashboard providing real-time site intelligence, precursor distributions, and longitudinal risk trends.

- **Executive KPI Stat Cards:** Total Reports Logged, SIF Precursors Detected, Critical Incidents (Score ≥ 80), High/Medium Priorities, Near Misses, and Pending HSE Reviews.
- **High-Risk Locations Ranking:** Ranks facilities and operating zones by aggregate SIF score and incident count with proportional progress bars.
- **Precursor Hazard Distribution Chart:** Visualizes frequency of underlying hazards (energized systems, falls from height, rotating machinery).
- **Critical Barrier Breakdown:** Highlights control failures (e.g., missing PPE, bypassed Lockout/Tagout, skipped gas testing).
- **Sequential Risk Score Trend:** Rolling chronological line chart to spot site safety deterioration before accidents occur.

Page 3: OSHA 2015–2025 AI Telemetry & Model Insights (URL: /model)

A full transparency dashboard detailing model performance on 105,996 severe workplace injury records from OSHA (2015–2025).

- **Confusion Matrix:** 15,445 True Positives, 4,907 True Negatives, 449 False Positives, and only 398 False Negatives (1.8% missed).
- **Learned Feature Weights Dictionary:** Searchable table of risk weights (fractured +24.5, amputated +22.8, broken +21.9, burns +15.3, concussion +13.6).
- **Live Model Playground:** Allows safety managers to type custom narratives and observe instant probability scores and word-level attributions.

Page 4: Alerts & Human-in-the-Loop Action Hub (URL: /alerts)

Centralized review queue for high-priority incidents where priority score reaches or exceeds **80 / 100**.

- **1-Click Action Buttons:** ✓ *Accept & Mitigate*, ■ *Escalate to Leadership* (with prompt modal for notes), and ✕ *Dismiss*.
- **Real-time Status Tabs:** Filter between Pending Action, Accepted, Escalated, and Dismissed.
- **Header Badge Synchronization:** Topbar displays an active red indicator with real-time pending alert count.

Page 5: Safety Report Log (URL: /reports)

Complete historical incident repository with search, filtering, and export capabilities.

- **Configurable Pagination:** Choose between 10, 25, or 50 records per page with next/previous navigation controls.
- **1-Click CSV & JSON Export:** Export reports formatted for Excel, PowerBI, or enterprise EHS systems.
- **Deep Inspection Modal:** Clicking any incident card opens an inspection drawer showing full narrative, priority score, and extracted precursors.

Page 6: Database & Cloud Architecture (MongoDB + SQLite)

Resilient dual-engine architecture: syncs to MongoDB Atlas (kanishkskct_db_user) with zero-downtime local SQLite fallback.

Summary for Administrators:

- Model runs locally on CPU in <15ms with zero cloud API costs.
- Use kanishkscet_db_user credentials in backend/.env.
- The TTL partial index maintains database storage permanently below the 512 MB free quota.